AWS Lab: EC2 Instance Family Sizing & Workload Profile Matching

Duration : 30 Min.

Scenario

A development team requests an expensive memory-optimized instance (r6i.4xlarge) for a batch image transcoding worker. The worker's CPU utilization is at 100% while memory usage remains under 8%. The cloud engineer must evaluate instance type families (General Purpose, Compute Optimized, Memory Optimized, Storage Optimized) and recommend a right-sized Compute-Optimized (c6i) instance family.

Input Details

The cloud environment has been pre-configured with the following parameters: - **Platform:** Amazon Web Services (EC2 Architecture) - **Input Metrics Telemetry:** `/opt/aws_data/workload_metrics.json` (CPU: ~100%, RAM: <8%) - **Evaluated Families:** T4g, M6i, C6i, R6i - **Target Architecture Fit:** Compute-Optimized (C6i family)

Username & Naming Conventions

Your candidate prefix is: **`<prefix>`** (derived from your login email before the `@` symbol, e.g. `student-demo01`). All custom resources and cloud configurations should incorporate this convention: - **Resource Naming:** Prepend or append `<prefix>` to your created AWS resources (e.g. `<prefix>-resource`). - **Candidate Tag:** Add tag `CandidatePrefix` = `<prefix>` on all managed cloud resources.

Task Objectives

Perform the following technical objectives directly in the AWS Management Console or using the AWS CLI: 1. Ingest resource utilization telemetry from /opt/aws_data/workload_metrics.json. 2. Compare resource ratios (vCPU to Memory) across T4g, M6i, C6i, and R6i instance families. 3. Evaluate and verify target resource sizing in the AWS console

Verification

Once completed, run the verification script to check your progress and receive your score. The verification system will inspect the live resources in the AWS cloud:

```
python run.py
```

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	Telemetry analysis: recognize high CPU and low memory profile from workload logs	4 Marks
TC2	Family selection: select Compute Optimized (C-family) as the ideal architecture fit	3 Marks
TC3	Recommendation report: JSON output contains exact instance type and cost delta	3 Marks

Total Score: 10 Marks

Important Notes

- Perform all actions directly in the AWS Management Console or using the AWS CLI.
- Ensure all resource configurations, tags, and naming conventions match the requirements.
- Do not terminate or alter pre-existing cloud infrastructure outside the scope of this assessment.